A Prefix-Rank Reformulation of the Sparse Limit in the Ordinal Matroid Secretary Problem

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Abstract

This note records a rigorous version of the valid part of a proposed solution to the matroidal open problem suggested by Correa, Cristi, Epstein, and Soto. The conclusion is deliberately modest. The argument below proves an exact equivalence between the ordinal adversarial-weight p-sample matroid secretary value and a newly defined prefix-rank contention-resolution value. Thus it is a correct reformulation of the same ordinal problem. It is not, by itself, a solution of the stronger external problem of identifying the limiting value with a standard matroid prophet, prophet-secretary, random-assignment, or i.i.d. model, nor does it prove that the limiting value is positive. In this sense the result is partial progress rather than a resolution of the authors' open problem.

1 Status of the result

The main mathematical point is the following. In a matroid, adversarial nonnegative weights can be decomposed into nonnegative gaps along the weight order. For a fixed online policy and a fixed hidden order, the expected weight obtained by the policy is therefore a nonnegative linear combination of its expected intersections with order prefixes, while the expected offline optimum is the same linear combination applied to the expected ranks of the active prefixes. Consequently the worst weight vector concentrates on the prefix at which the ratio

$$\frac{\mathbb{E}[|A \cap P|]}{\mathbb{E}[r(R \cap P)]}$$

is smallest.

This gives a clean equivalence, but the equivalence should not be overinterpreted. The prefix-rank model defined below is designed to have exactly the same information structure as the ordinal p-sample problem and an objective obtained by dualizing adversarial weights. Therefore equality of the two values is a theorem, but it is not evidence that the sparse limit equals the value of any pre-existing cardinal-information matroid prophet inequality. The result should be advertised as an exact reformulation and a diagnostic tool.

2 The ordinal p-sample value

Let $M = (E, \mathcal{I})$ be a finite matroid with rank function r_M, and write $n = |E|$. We assume $r_M(E) > 0$; rank-zero matroids have zero offline benchmark for every weight vector and may be treated by any harmless convention. Fix $q \in (0, 1]$ and put $p = 1 - q$.

Nature draws an active set $R \subseteq E$ by including every element independently with probability q. The set $H = E \setminus R$ is the history set. The active elements of R subsequently arrive in uniformly random order. A nonanticipatory online policy first sees H together with the induced relative order of the elements of H. When an active element arrives, the policy learns its relative position among all elements revealed so far, and must irrevocably accept or reject it while maintaining an independent selected set.

The word "ordinal" is important: throughout this note, policies do not see the numerical weights and cannot condition on cardinal gaps between consecutive weights. They see only the matroid, the thinning parameter, the history sample, the arrival order, and relative-order information.

For a nonnegative weight function $w : E \to \mathbb{R}_+$, let

$$\mathrm{OPT}_q(M, w) := \mathbb{E}_R \left[\max \left\{ \sum_{e \in I} w(e) : I \in \mathcal{I},\ I \subseteq R \right\} \right]. \tag{1}$$

Weight functions with $\mathrm{OPT}_q(M, w) = 0$ are ignored in competitive-ratio infima, since both the online and offline values are then zero on all non-loop elements of positive value. Let A_Π be the output of policy Π, including all randomness from thinning, arrivals, and the policy.

Let $\mathfrak{A}(M, q)$ denote the set of admissible ordinal policies. The ordinal adversarial-weight p-sample value is

$$\beta_M^{\mathrm{ord}}(1-q) := \sup_{\Pi \in \mathfrak{A}(M,q)} \inf_{\substack{w: E \to \mathbb{R}_+ \\ \mathrm{OPT}_q(M,w) > 0}} \frac{\mathbb{E}[w(A_\Pi)]}{\mathrm{OPT}_q(M, w)}. \tag{2}$$

For a class $\mathcal{C}$ of finite matroids of positive rank, set

$$\beta_{\mathcal{C}}^{\mathrm{ord}}(1-q) := \inf_{M \in \mathcal{C}} \beta_M^{\mathrm{ord}}(1-q). \tag{3}$$

If the symbol β_M in a source is intended to range over stronger policies that observe cardinal weights, then (2) is a restriction of that value. The equivalence proved here concerns the ordinal value (2).

3 The prefix-rank contention-resolution value

A *maximal chain* of E is a sequence

$$\mathcal{P} = (P_0, P_1, \ldots, P_n), \qquad \emptyset = P_0 \subset P_1 \subset \cdots \subset P_n = E, \qquad |P_i| = i.$$

We write $P_i = \{e_1, \ldots, e_i\}$, so that the chain is equivalently a total order

$$e_1 \succ e_2 \succ \cdots \succ e_n.$$

The intended interpretation is that P_i consists of the i heaviest elements, with ties resolved by a fixed deterministic rule.

Definition 3.1 (Sparse ordinal prefix-rank contention resolution)**.** *Fix M, q, and a hidden chain $\mathcal{P}$. The stochastic process and the policy's observations are exactly as in the ordinal p-sample model: $R = E[q]$ is generated by independent thinning, $H = E \setminus R$ is observed as history with its induced order, active elements arrive in random order, and the policy sees only relative-order information. For a policy $\Pi \in \mathfrak{A}(M, q)$, define*

$$a_i(\Pi, M, \mathcal{P}, q) := \mathbb{E}[|A_\Pi(\mathcal{P}) \cap P_i|], \tag{4}$$

$$b_i(M, \mathcal{P}, q) := \mathbb{E}[r_M(R \cap P_i)]. \tag{5}$$

The prefix-rank balance of Π *against* $\mathcal{P}$ *is*

$$\gamma(\Pi; M, \mathcal{P}, q) := \min_{\substack{1\le i\le n\\ b_i(M,\mathcal{P},q)>0}} \frac{a_i(\Pi, M, \mathcal{P}, q)}{b_i(M, \mathcal{P}, q)}. \tag{6}$$

The value on M *is*

$$\rho_M(q) := \sup_{\Pi\in\mathfrak{A}(M,q)} \inf_{\mathcal{P}} \gamma(\Pi; M, \mathcal{P}, q), \tag{7}$$

and for a class $\mathcal{C}$*,*

$$\rho_{\mathcal{C}}(q) := \inf_{M\in\mathcal{C}} \rho_M(q). \tag{8}$$

Equivalently, a policy is c-balanced against $\mathcal{P}$ if

$$\mathbb{E}[|A_\Pi(\mathcal{P}) \cap P_i|] \ge c\,\mathbb{E}[r_M(R \cap P_i)] \qquad \text{for every } i = 0, 1, \dots, n. \tag{9}$$

The condition at $i = 0$ is void. Prefixes with zero expected active rank are also void: if $b_i = 0$, then P_i contains only loops, so no independent selected set contains an element of P_i.

4 The decomposition lemma

The proof rests on the standard matroid greedy theorem, combined with a telescoping identity.

Lemma 4.1 (Prefix decomposition)**.** *Let* $M = (E, \mathcal{I})$ *be a matroid. Order the elements as*

$$e_1, e_2, \dots, e_n, \qquad w(e_1) \ge w(e_2) \ge \cdots \ge w(e_n) \ge 0,$$

with ties resolved consistently with the displayed order. Put $P_i = \{e_1, \dots, e_i\}$*,* $w(e_{n+1}) = 0$*, and*

$$\Delta_i := w(e_i) - w(e_{i+1}) \ge 0.$$

Then:

(i) for every set $A \subseteq E$*,*

$$w(A) = \sum_{i=1}^{n} \Delta_i |A \cap P_i|; \tag{10}$$

(ii) for every $R \subseteq E$*,*

$$\max\{w(I) : I \in \mathcal{I},\ I \subseteq R\} = \sum_{i=1}^{n} \Delta_i r_M(R \cap P_i). \tag{11}$$

Proof. For every j,

$$w(e_j) = \sum_{i=j}^{n} \Delta_i.$$

Therefore

$$w(A) = \sum_{e_j\in A} \sum_{i=j}^{n} \Delta_i = \sum_{i=1}^{n} \Delta_i |A \cap P_i|,$$

which proves (10).

For (11), run the greedy algorithm on the set R in the order $e_1, e_2, \ldots, e_n$, accepting an element precisely when doing so preserves independence. By the matroid greedy theorem, the resulting set G is a maximum-weight independent subset of R. Moreover, after the greedy algorithm has scanned the prefix P_i, its current set is a basis of $R \cap P_i$; hence

$$|G \cap P_i| = r_M(R \cap P_i) \qquad \text{for every } i.$$

Applying (10) to G gives

$$w(G) = \sum_{i=1}^{n} \Delta_i |G \cap P_i| = \sum_{i=1}^{n} \Delta_i r_M(R \cap P_i),$$

and $w(G)$ is the offline optimum on R. $\square$

5 Fixed-order equivalence

For a fixed chain $\mathcal{P}$, let $\mathcal{W}(\mathcal{P})$ be the set of all nonnegative weight functions whose weakly decreasing order, after the fixed tie-breaking rule, is $e_1 \succ \cdots \succ e_n$.

Proposition 5.1 (Worst weights are worst prefixes)**.** *Fix M, q, a policy $\Pi \in \mathfrak{A}(M, q)$, and a chain $\mathcal{P}$. Then*

$$\inf_{\substack{w \in \mathcal{W}(\mathcal{P}) \\ \mathrm{OPT}_q(M,w)>0}} \frac{\mathbb{E}[w(A_\Pi(\mathcal{P}))]}{\mathrm{OPT}_q(M, w)} = \gamma(\Pi; M, \mathcal{P}, q). \tag{12}$$

The infimum need not be attained if the model requires a strict weight order.

Proof. Write $a_i = a_i(\Pi, M, \mathcal{P}, q)$ and $b_i = b_i(M, \mathcal{P}, q)$. For $w \in \mathcal{W}(\mathcal{P})$, let $\Delta_i = w(e_i) - w(e_{i+1}) \geq 0$ be its gaps. Taking expectations in Theorem 4.1 gives

$$\mathbb{E}[w(A_\Pi(\mathcal{P}))] = \sum_{i=1}^{n} \Delta_i a_i, \tag{13}$$

$$\mathrm{OPT}_q(M, w) = \sum_{i=1}^{n} \Delta_i b_i. \tag{14}$$

If $b_i = 0$, then $R \cap P_i$ has rank zero almost surely; since $q > 0$, this means P_i has rank zero. Thus P_i consists only of loops, and an independent selected set cannot intersect P_i. Hence also $a_i = 0$. Such indices contribute neither to (13) nor to (14).

Assume $\mathrm{OPT}_q(M, w) > 0$. Then

$$\frac{\mathbb{E}[w(A_\Pi(\mathcal{P}))]}{\mathrm{OPT}_q(M, w)} = \frac{\sum_i \Delta_i a_i}{\sum_i \Delta_i b_i} = \sum_{i:b_i>0} \lambda_i \frac{a_i}{b_i}, \qquad \lambda_i := \frac{\Delta_i b_i}{\sum_j \Delta_j b_j}.$$

The numbers λ_i are nonnegative and sum to 1. Hence the ratio is at least

$$\min_{i:b_i>0} \frac{a_i}{b_i} = \gamma(\Pi; M, \mathcal{P}, q).$$

Conversely, let k attain the above minimum. For $\varepsilon > 0$, define

$$w_k^\varepsilon(e_j) = \mathbf{1}\{j \leq k\} + \varepsilon(n - j), \qquad j = 1, \ldots, n.$$

This weight vector is strictly decreasing in the order $e_1 \succ \cdots \succ e_n$, and hence belongs to $\mathcal{W}(\mathcal{P})$ regardless of how ties would otherwise be handled. Since $b_k > 0$, its offline benchmark is positive for all sufficiently small ε. As $\varepsilon \downarrow 0$, the gap vector of w_k^ε converges to the vector with $\Delta_k = 1$ and $\Delta_i = 0$ for $i \neq k$. The expectations in (13) and (14) are finite linear functions of the gaps, so the corresponding ratio converges to $a_k/b_k = \gamma(\Pi; M, \mathcal{P}, q)$. $\square$

6 Equivalence of values

Theorem 6.1 (Ordinal p-sample value equals prefix-rank value)**.** *For every finite matroid M of positive rank and every $q \in (0,1]$,*

$$\beta_M^{\mathrm{ord}}(1-q) = \rho_M(q). \tag{15}$$

Consequently, for every class $\mathcal{C}$ of finite positive-rank matroids,

$$\beta_{\mathcal{C}}^{\mathrm{ord}}(1-q) = \rho_{\mathcal{C}}(q). \tag{16}$$

Proof. Fix M, q, and a policy Π. Every nonnegative weight function with positive offline benchmark induces a maximal chain after tie-breaking. Conversely, for every maximal chain $\mathcal{P}$, the class $\mathcal{W}(\mathcal{P})$ consists exactly of the nonnegative weight functions inducing that chain. Since the policy is ordinal, its behavior for weights in $\mathcal{W}(\mathcal{P})$ depends on the chain and the revealed relative-order process, not on the cardinal gaps of the weights.

Therefore

$$\begin{aligned}\inf_{\substack{w:E\to\mathbb{R}_+\\ \mathrm{OPT}_q(M,w)>0}} \frac{\mathbb{E}[w(A_\Pi)]}{\mathrm{OPT}_q(M,w)} &= \inf_{\mathcal{P}} \inf_{\substack{w\in\mathcal{W}(\mathcal{P})\\ \mathrm{OPT}_q(M,w)>0}} \frac{\mathbb{E}[w(A_\Pi(\mathcal{P}))]}{\mathrm{OPT}_q(M,w)} \\ &= \inf_{\mathcal{P}} \gamma(\Pi; M, \mathcal{P}, q),\end{aligned}$$

where the second equality is Theorem 5.1. Taking the supremum over $\Pi \in \mathfrak{A}(M,q)$ proves (15). Taking the infimum over $M \in \mathcal{C}$ proves (16). $\square$

Corollary 6.2 (Sparse limiting identity)**.** *If the relevant limits exist, then*

$$\lim_{p\uparrow 1} \beta_{\mathcal{C}}^{\mathrm{ord}}(p) = \lim_{q\downarrow 0} \rho_{\mathcal{C}}(q). \tag{17}$$

The same statement holds with both limits replaced by liminf, and also with both replaced by limsup.

Proof. Substitute $q = 1-p$ in Theorem 6.1. $\square$

7 Rank-one specialization

Let M have rank one. Then for every prefix P_i,

$$r_M(R\cap P_i) = \mathbf{1}\{R\cap P_i \neq \emptyset\}.$$

Every feasible selected set has size at most one, so

$$|A_\Pi(\mathcal{P})\cap P_i| = \mathbf{1}\{A_\Pi(\mathcal{P})\cap P_i \neq \emptyset\}.$$

Thus the prefix-rank condition (9) becomes

$$\mathbb{P}[A_\Pi(\mathcal{P})\cap P_i \neq \emptyset] \geq c\,\mathbb{P}[R\cap P_i \neq \emptyset] \qquad \text{for every prefix } P_i. \tag{18}$$

This is the usual stochastic-dominance form of the single-choice adversarial-values problem: for every threshold prefix, the gambler's selected item lies in that prefix with probability at least a c fraction of the prophet's probability of having an active item in the same prefix.

Consequently, in rank one the prefix-rank formulation is not an exotic new condition; it is exactly the standard threshold-prefix condition. In higher-rank matroids, the event "the active maximum is in the prefix" is replaced by the random rank $r_M(R\cap P_i)$.

8 What this does and does not solve

The theorem proves that the prefix-rank game is an exact ordinal dual formulation of the adversarial-weight p-sample matroid secretary problem. Standard neighboring models include matroid prophet inequalities, random-assignment matroid secretary, and contention-resolution formulations of matroid secretary [3, 4, 2]. The present reformulation is useful for at least two reasons.

1. It removes cardinal weights from the statement. Any lower bound for $\rho_M(q)$ is automatically a lower bound for the ordinal adversarial-weight value $\beta_M^{\text{ord}}(1-q)$, and any upper bound can be witnessed by a single bad prefix.

2. It identifies the natural family of constraints that an ordinal sparse-limit algorithm must satisfy: simultaneously for every hidden weight prefix P, it must capture a constant fraction of $\mathbb{E}[r_M(R \cap P)]$ in expectation.

However, the theorem does not settle the stronger interpretation of the open problem. In particular, it does not prove any of the following statements:

1. that $\lim_{q \downarrow 0} \rho_{\mathcal{C}}(q)$ is positive for the class of all matroids;

2. that the limiting value is equal to the optimal ratio of a standard i.i.d. matroid prophet inequality, matroid prophet-secretary problem, or random-assignment matroid secretary problem;

3. that the ordinary matroid secretary conjecture has a constant-competitive solution;

4. that algorithms using cardinal weight information have the same value as ordinal algorithms.

Thus the correct status is:

> The central proof contains no serious mathematical error as an equivalence theorem for the ordinal model. It gives partial progress by reformulating the value exactly as a prefix-rank contention-resolution problem. It is not a full solution of the authors' open problem unless that open problem is interpreted as allowing the analogue to be a newly defined problem whose objective is chosen to be exactly equivalent to the original adversarial-weight value.

References

[1] José Correa, Andrés Cristi, Boris Epstein, and José A. Soto. Sample-driven optimal stopping: From the secretary problem to the i.i.d. prophet inequality. *Mathematics of Operations Research*, 49(1):441–475, 2024.

[2] Shaddin Dughmi. Matroid secretary is equivalent to contention resolution. *Proceedings of ITCS 2022*, LIPIcs 215:58:1–58:23, 2022.

[3] Robert Kleinberg and S. Matthew Weinberg. Matroid prophet inequalities. *Proceedings of STOC 2012*, pages 123–136, 2012.

[4] José A. Soto. Matroid secretary problem in the random-assignment model. *SIAM Journal on Computing*, 42(1):178–211, 2013.